

RIPHAH INTERNATIONAL UNIVERSITY



Faculty of Computing

FINAL YEAR PROJECT INITIAL PROPOSAL

Deep Forensics - Multimodal Deepfake Detection System for Bank Video-KYC

Project Team

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1. Description:

Deep Forensics is a self-hosted, web-based verification platform designed to secure **bank video-KYC** and remote onboarding sessions by jointly detecting facial deepfakes and AI-cloned audio. Combining pre-trained computer vision and speech classification models with Grad-CAM visual explainability, it delivers an auditable, cryptographically hashed forensic report with a clear authenticity verdict (Real, Suspicious, or Fake) for compliance officers.

2. Opportunity & Stakeholders

Banks and fintechs are rapidly shifting customer onboarding and high-value transaction approval to remote video-KYC (Know Your Customer) calls, in line with the State Bank of Pakistan's digital onboarding initiatives and global video-KYC adoption. The live video call has effectively become the new front door to a bank account, and it is now a prime target for fraud.

The **primary stakeholder** is the **bank's KYC** / identity-verification team, the compliance officers and **onboarding agents** who must decide, in real time, whether the face and voice presented on a call belong to a genuine, live human being. Secondary stakeholders include the bank's fraud and risk department, the end customer (protected from impersonation), and the regulator (who requires an auditable verification trail). A further stakeholder group is the corporate sector, where **HR and hiring teams** can use the same system during remote interviews and onboarding to verify that a new candidate is a genuine, live person, extending the tool's use case from bank video-KYC to corporate candidate verification.

3. Existing Systems

Current practice relies largely on manual agent review, yet humans are demonstrably poor at detecting modern AI-generated forgeries. Commercial liveness and KYC vendors (such as **Onfido**, **iProov**, and **Jumio**) are effective but are closed-source, **costly**, foreign-hosted, and raise data-residency concerns that make them **difficult for smaller local banks** to adopt. Most academic deepfake detectors, meanwhile, check either the face or the voice in isolation, output only a raw probability score, and provide no auditable report that a compliance officer could act upon. The gap is the absence of an accessible, self-hostable tool that jointly analyses both the visual and audio channels of a KYC session and produces a clear, auditable forensic verdict.

4. Problem Statement

Remote video-KYC sessions are increasingly attacked using face-swap deepfakes and AI voice-cloning, yet bank KYC teams lack an affordable, self-hosted system that jointly analyses the visual and audio channels of a session and delivers a clear, auditable authenticity decision.

5. Proposed Solution

We propose Deep Forensics, a web-based multimodal deepfake forensic system in which a KYC officer uploads a short clip captured from a verification session and receives a combined authenticity report. The system integrates existing, pre-trained detection models as components (used for inference, with light fine-tuning on Google Colab where required); it does not train models from scratch and does not attempt novel model research. It comprises the following components:

5.1 - Module A - Face / Image Forensics: extracts video frames, detects the face, and applies a pre-trained deepfake-image classifier (for example, an **EfficientNet**- or **Xception**-based model pre-trained on FaceForensics++ / DFDC) to score visual manipulation, together with a Grad-CAM explainability heatmap indicating which facial region appears manipulated.

5.2 - Module B - Audio Forensics: extracts the audio track, computes **mel-spectrogram** features and pre-trained synthetic-speech-detection embeddings and scores the likelihood of voice cloning or synthesis.

5.3 - Fusion & Decision: combines the two module outputs through a transparent, rule-based risk score into a single verdict of **REAL**, **SUSPICIOUS**, or **FAKE**, with configurable thresholds. The fusion is deliberately rule-based rather than a learned architecture, keeping the project a software solution rather than a research contribution.

5.4 - Reporting & User Interface: a clean **web dashboard** (upload, analyse, verdict) and a downloadable PDF forensic report containing the scores, heatmap, timestamp, and a content hash for the audit trail.

6. Project Scope

6.1 - In scope: face-based deepfake detection using a pre-trained model; audio synthetic-voice detection using a pre-trained model; rule-based fusion into a single KYC verdict; a web application for upload, analysis, and results; visual explainability and a downloadable PDF forensic report; and evaluation on public benchmark datasets (FaceForensics++ / DFDC subsets for video, and an ASVspoof subset for audio).

6.2 - Out of scope: real-time live-stream analysis (the system operates on uploaded clips in batch / near-real-time); training deep models from scratch or developing novel fusion architectures; integration with any production banking system; and a mobile application (the deliverable is web-only).

7. Project Timeline

The project will be completed over two academic semesters (approximately 32 weeks in total), with a progress update delivered every four weeks as shown below.

Table1.1: Project road map

Semester	Weeks	Milestone / Update
Semester 1	Weeks 1-4	Literature review, requirement analysis, and collection/preparation of benchmark datasets (FaceForensics++ / DFDC, ASVspoof).
Semester 1	Weeks 5-8	Development of Module A (Face / Image Forensics): face detection pipeline and integration of the pre-trained deepfake-image classifier.
Semester 1	Weeks 9-12	Development of Module B (Audio Forensics): audio extraction, mel-spectrogram feature generation, and synthetic-speech-detection scoring.
Semester 1	Weeks 13-16	Unit testing of Modules A and B on sample clips; mid-semester evaluation and FYP-I documentation/submission.
Semester 2	Weeks 17-20	Design and implementation of the rule-based Fusion & Decision module producing the REAL / SUSPICIOUS / FAKE verdict.
Semester 2	Weeks 21-24	Development of the web dashboard for upload, analysis, and results display.
Semester 2	Weeks 25-28	Integration of the Grad-CAM explainability heatmap and generation of the downloadable, auditable PDF forensic report.
Semester 2	Weeks 29-32	System-wide testing and evaluation on benchmark datasets, bug fixing, final documentation, and FYP-II submission/defense.

8. Final Product

At the end of the project, Deep Forensics will be delivered as a fully **functional**, self-hostable web application. A bank KYC officer or a **corporate HR team** will be able to upload a short video clip from a verification session and receive, within a few minutes, a combined authenticity report. The final product will include: a working web dashboard for uploading and analysing clips; the integrated face-forensics and audio-forensics modules producing a single REAL / SUSPICIOUS / FAKE verdict through rule-based fusion; a Grad-CAM visual explainability heatmap highlighting the facial region flagged as manipulated; a downloadable, auditable PDF forensic report containing the scores, heatmap, timestamp, and a content hash; and an evaluation report documenting the system's accuracy on public benchmark datasets (FaceForensics++ / DFDC and ASVspoof). The deliverable will be a web-only, self-hosted proof-of-concept system, not a production banking integration.